

Longitude and Latitude - Finding the Way at Sea

UK Family Tour

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Standing beneath the copper hull of the Cutty Sark this morning, it is worth remembering that this ship was, above everything else, a machine for crossing empty ocean and arriving exactly where she meant to. A tea clipper that could not find her way was worth nothing. To understand how her captains threaded ten thousand miles of featureless water from Shanghai to London, you have to understand the two invisible lines that every navigator lives by, and why one of them was easy and the other very nearly impossible.

Latitude, the north-south measurement, came first and came easily, because nature hands it to you. Stand on deck at noon, measure how high the sun climbs above the horizon, correct for the day of the year, and you know how far you are from the equator. At night you can do the same with the Pole Star, which sits almost exactly above the North Pole and barely moves. The Greeks knew this. Ptolemy was drawing lines of latitude across his maps in the second century. A sailor in 1500, or in 1869, could fix his latitude within a few miles using a sextant and a table, and the tool for it, measuring the angle between a star and the horizon, is one of the loveliest instruments ever made.

Longitude, the east-west measurement, was a nightmare. There is no east-west equivalent of the Pole Star, nothing fixed in the sky to sight against, because the whole heavens wheel overhead as the Earth turns. And here is the crucial insight the whole problem turns on: longitude is really a measurement of time. The Earth spins through three hundred and sixty degrees in twenty-four hours, which means fifteen degrees of longitude every hour. So if you know what time it is where you are, by looking at your own local noon when the sun is highest, and you also know at that exact instant what time it is back at your home port, the difference between the two clocks tells you how far east or west you have sailed. One hour of difference is fifteen degrees. Four minutes is one degree. The mathematics is simple. The engineering was murderous.

The trouble was the second clock, the one that had to keep home-port time faithfully through months at sea. No

clock on Earth could do it. Pendulum clocks, the most accurate timekeepers on land, were useless on a heaving deck, where the swing of the pendulum was ruined by every roll and pitch. Temperature swung from tropics to Cape Horn and changed the length of every metal part. Salt air corroded, damp swelled the oil. A clock that lost or gained even a few seconds a day would, over a two-month voyage, put a ship dozens or hundreds of miles from where it thought it was, and ships died of that error. In 1707 a British fleet, confident of its position and wrong about it, piled onto the rocks of the Isles of Scilly on a foggy October night and around fourteen hundred to two thousand sailors drowned. The nation was appalled.

So in 1714 Parliament passed the Longitude Act and offered a fortune, up to twenty thousand pounds, the equivalent of millions today, to anyone who could find a reliable way to determine longitude at sea. Two camps formed, and their rivalry is one of the great stories of science. On one side stood the astronomers, who believed the answer was in the sky. Their method, called lunar distance, used the Moon as a clock: the Moon moves against the background stars at a steady, predictable rate, so by measuring the angle between the Moon and a chosen star and consulting elaborate pre-computed tables, a navigator could work out the time at Greenwich and therefore his longitude. It worked, but it demanded a clear sky, a good view of the Moon, and around four hours of fearsome calculation for a single fix.

On the other side stood a Yorkshire carpenter and self-taught clockmaker named John Harrison, who thought the astronomers had it backwards. The answer was not to read the sky but to build a clock so perfect it could carry home-port time in a box. Over forty years he built a series of extraordinary machines, and if you climb the hill to the Royal Observatory this afternoon you can see them still ticking. His first three, H1, H2 and H3, were large and brilliant and not quite good enough. Then in 1759 he produced H4, which looks like an oversized pocket watch, and it changed everything. On a voyage to Jamaica it lost only about five seconds over eighty-one days, an accuracy no one had believed

possible.

Harrison should have been hailed and paid at once. Instead the Board of Longitude, stacked with astronomers who favoured the lunar method, and in particular the Astronomer Royal Nevil Maskelyne, who was Harrison's rival and also one of his judges, moved the goalposts again and again. They demanded he hand over his designs, build copies, submit to more trials. Harrison, by now an old man, was nearly cheated of his reward, and only the personal intervention of King George the Third, who reportedly declared "By God, Harrison, I will see you righted," got him most of his money through a special act of Parliament in 1773. He died three years later. The clock had won, but only just.

The verdict of history came from Captain James Cook, who carried a copy of Harrison's watch, made by Larcum Kendall and called K1, on his second great Pacific voyage in the 1770s. Cook called it his "trusty friend" and "never-failing guide," and used it to chart coastlines with an accuracy that had never been seen. Within a few decades, as chronometers became cheaper and more numerous, every serious ship carried one or several. By the time *Cutty Sark* was launched in 1869 the marine chronometer was simply part of the furniture, kept below in gimbals to stay level, wound with ritual care, checked against time signals. Every day at sea her officers took a noon sight of the sun for latitude, compared their chronometer to work out longitude, and pricked the ship's position onto the chart. That daily ceremony, sextant and chronometer and a moment of arithmetic, was how she won her races.

And this brings us to why the world's clocks are set to this particular hill in south-east London. If longitude is the difference between your local time and the time at a reference meridian, then everyone has to agree on which meridian counts as zero. For a long time they did not. The French measured from Paris, others from their own capitals, and charts from different nations disagreed. The Royal Observatory at Greenwich had been founded in 1675 by King

Charles the Second for exactly this purpose, "to find the so-much desired longitude of places for the perfecting the art of navigation," and generations of British astronomers had produced the star catalogues and almanacs that made the lunar method work. So many of the world's ships were already using charts based on Greenwich that in 1884, at a conference in Washington attended by twenty-five nations, Greenwich was voted the Prime Meridian of the world, longitude zero. The French abstained and sulked, keeping Paris time under a euphemism until 1911. But the decision held, and it is why the brass line you will straddle in the Observatory courtyard, one foot in the eastern hemisphere and one in the western, is the line from which the entire planet measures east and west.

There is a lovely modern footnote to stand on that line for. Satellite navigation, which now does instantly and invisibly what took Harrison forty years to make possible, does not quite agree with the old brass strip. The GPS zero of longitude sits about a hundred metres to the east, out across the courtyard and into the park, because the satellites reference a mathematical model of the whole Earth rather than the local vertical that the Victorian astronomers used with their telescopes. So there are, in a sense, two prime meridians a hundred metres apart, and you can walk from one age of navigation to the next in about ninety seconds.

That is the real cargo of this small hill. *Cutty Sark* carried tea and wool, but Greenwich exported something less tangible and far more valuable: agreement. The grid of latitude and longitude that lets any point on Earth be named by two numbers, the reference meridian that lets those numbers mean the same thing to everyone, the fusion of clock and star that turned a deadly guessing game into a routine, these are the quiet infrastructure of a connected world. When your phone tells you where you are tonight, it is finishing a sentence that a drowned fleet, a stubborn carpenter, and a racing clipper all helped to write.